Object Oriented Programming (CSEG1044)

Unit 06 – Lecture 01: Capstone Kickoff + Team Setup + Git/GitHub

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February 27, 2026

Sources: Git SCM: *Pro Git* (2e) (version control workflow), accessed February 27, 2026.; GitHub Docs (repositories, issues, pull requests), accessed February 27, 2026.; Oracle Java Tutorials: Swing + JDBC (accessed February 27, 2026).; JUnit 5 User Guide (accessed February 27, 2026).; Apache Maven Project: Getting Started (accessed February 27, 2026).; Gradle User Manual: Getting Started (accessed February 27, 2026).; UML class diagrams: UML 2.x overview/spec (accessed February 27, 2026).

Quick Links

Overview Key Topics Demo Summary

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Agenda

1 Overview

2 Key Topics

3 Demo / Example

4 Summary

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Learning Outcomes

- Select a project theme with clear scope boundaries.

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Learning Outcomes

- Select a project theme with clear scope boundaries.
- Set up team repo and workflow (issues, branches, PRs).

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Learning Outcomes

- Select a project theme with clear scope boundaries.
- Set up team repo and workflow (issues, branches, PRs).
- Complete Deliverable-0: proposal + initial backlog.

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Unit 06: Capstone Project

- Lecture focus: Capstone Kickoff + Team Setup + Git/GitHub

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Today: Roadmap

- Problem statement options, scope boundaries, evaluation criteria

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Today: Roadmap

- Problem statement options, scope boundaries, evaluation criteria
- Team setup, repo creation, issues, branching conventions

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Today: Roadmap

- Problem statement options, scope boundaries, evaluation criteria
- Team setup, repo creation, issues, branching conventions
- Deliverable-0: project proposal (1 page) + initial backlog

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Capstone goal (what you will build)

- A small complete product: Swing UI + JDBC + OOP design.

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- A small complete product: Swing UI + JDBC + OOP design.
- Evidence of teamwork: GitHub issues, branches, PRs.

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Capstone goal (what you will build)

- A small complete product: Swing UI + JDBC + OOP design.
- Evidence of teamwork: GitHub issues, branches, PRs.
- Clear scope and deliverables across Unit 06 lectures.

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Scope boundaries (must do)

- Must-have: required for minimum demo.

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Scope boundaries (must do)

- Must-have: required for minimum demo.
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- Must-have: required for minimum demo.
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Scope boundaries (must do)

- Must-have: required for minimum demo.
- Should-have: good improvements if time permits.
- Could-have: optional enhancements.
- Non-goals: explicitly out of scope.

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Team setup checklist

1. Create repo + add collaborators

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Team setup checklist

1. Create repo + add collaborators
2. Add README (project + run steps + team)

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Team setup checklist

1. Create repo + add collaborators
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3. Create labels (feature, bug, ui, db, doc)

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Team setup checklist

1. Create repo + add collaborators
2. Add README (project + run steps + team)
3. Create labels (feature, bug, ui, db, doc)
4. Create initial issues (8–12)

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Suggested repo skeleton (keep it simple)

```
repo-root/
  README.md
  docs/ (deliverables: D0, D1, D2...)
  src/ (Java packages)
  db/ (schema.sql)
```

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Git workflow (simple)

- Keep `main` stable.

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- Keep `main` stable.
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Git workflow (simple)

- Keep `main` stable.
- Work on feature branches: `feature/<name>`.
- Open PRs, review, then merge.
- Use clear commit messages.

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What a good GitHub issue includes

- Clear goal (one feature or one bug fix)

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What a good GitHub issue includes

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- Clear goal (one feature or one bug fix)
- Acceptance criteria (3–6 testable bullets)
- Task checklist (UI/service/DAO/DB/test/doc)
- Label + assignee + link to PR (when created)

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Deliverable-0 checklist

- 1-page proposal: title, problem, Must/Should/Could, screens, DB tables, milestones, risks, repo link.

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Deliverable-0 checklist

- 1-page proposal: title, problem, Must/Should/Could, screens, DB tables, milestones, risks, repo link.
- Backlog: issues for must-have features + technical tasks.
- Evidence: at least 1 PR created (even small setup PR).

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In-class Activity (evidence)

- Repo + collaborators + README

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In-class Activity (evidence)

- Repo + collaborators + README
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In-class Activity (evidence)

- Repo + collaborators + README
- Issues created (must-have + setup)
- One branch + one PR created

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Key Takeaways

- Define scope early to avoid project failure.

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Key Takeaways

- Define scope early to avoid project failure.
- Use GitHub workflow (issues + PRs) for collaboration.

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Key Takeaways

- Define scope early to avoid project failure.
- Use GitHub workflow (issues + PRs) for collaboration.
- Deliverable-0 sets the foundation for the entire capstone.

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Checkpoint Questions

1. Why define Must/Should/Could/Non-goals?

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Checkpoint Questions

1. Why define Must/Should/Could/Non-goals?
2. Why use branches + PRs?

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Checkpoint Questions

1. Why define Must/Should/Could/Non-goals?
2. Why use branches + PRs?
3. What is the minimum content of Deliverable-0?

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Next Lecture Preview

Next: Requirements → Use Cases → Milestones

- Convert the proposal into clear use cases and a working plan.

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